Complete Live Trading Implementation Checklist

🔧 STEP 1: Add New Safety Components (Do This Week)

A. Add Circuit Breaker System

```bash
# Create new file
mkdir -p src/ib_quant_kit/safety
touch src/ib_quant_kit/safety/__init__.py
```

- [ ] Copy `circuit_breaker.py` artifact to `src/ib_quant_kit/safety/`
- [ ] Add to `.env`:

```bash
MAX_DAILY_LOSS=1000.0
MAX_DRAWDOWN=2000.0
MAX_TOTAL_NOTIONAL=100000.0
```

- [ ] Test circuit breaker triggers correctly:

```python
from src.ib_quant_kit.safety.circuit_breaker import get_circuit_breaker
cb = get_circuit_breaker()
cb.initialize_session(100000.0)
# Simulate loss
ok, reason = cb.update_equity(98500.0)  # Should trigger if > $1000 loss
assert not ok  # Should fail
assert "DAILY LOSS" in reason
```

B. Add Commission Tracking

```bash
mkdir -p src/ib_quant_kit/costs
touch src/ib_quant_kit/costs/__init__.py
```

- [ ] Copy `commission.py` artifact to `src/ib_quant_kit/costs/`
- [ ] Integrate into `ib_client.py` `execDetails`:

```python
def execDetails(self, reqId, contract, execution):
    # ... existing code ...

    # Add cost tracking
    from .costs.commission import cost_tracker
    cost_tracker.record_fill(
        symbol=contract.symbol,
        side=execution.side,
        qty=float(execution.shares),
        price=float(execution.price),
        sec_type=contract.secType,
        order_type="LMT",  # or track actual order type
        actual_commission=execution.commission if hasattr(execution, 'commission') else None
    )
```

- [ ] Update PnL reporting to show net after costs:

```python
pnl_snap = pnl.snapshot()
cost_snap = cost_tracker.get_snapshot()
net_pnl = cost_tracker.get_net_pnl(pnl_snap['net'])
print(f"Gross PnL: ${pnl_snap['net']:.2f}")
print(f"Costs: ${cost_snap['total']:.2f}")
print(f"Net PnL: ${net_pnl:.2f}")
```

C. Add Order Lifecycle Management

```bash
mkdir -p src/ib_quant_kit/orders
touch src/ib_quant_kit/orders/__init__.py
```

- [ ] Copy `lifecycle.py` artifact to `src/ib_quant_kit/orders/`
- [ ] **DELETE** duplicate `src/ib_quant_kit/orders/order_book.py` (keep main `orders.py`)
- [ ] Integrate into `ib_client.py`:

```python
```

```python
# Add to IBClient class:

def cancelOrder(self, orderId: int):
    """Cancel an order"""
    super().cancelOrder(orderId)
    from .orders.lifecycle import order_lifecycle
    order_lifecycle.update_status(orderId, "PendingCancel")
    console.log(f"[yellow]Cancelling order {orderId}[/yellow]")

def submit_limit(self, contract, side, qty, limit, tif, outside_rth, idempotency_key):
    # Existing code...
    client_order_id = order_book.get_or_assign(idempotency_key, self.next_client_order_id())

    # Register with lifecycle manager
    from .orders.lifecycle import order_lifecycle
    order_lifecycle.register_order(
        order_id=client_order_id,
        symbol=contract.symbol,
        side=side,
        qty=qty,
        limit_price=limit,
        order_type="LMT",
        timeout_seconds=300  # 5 minute timeout
    )

    # ... rest of existing code
```

D. Update IBClient with Missing Methods

- [ ] Add `position()` callback (from earlier artifact)
- [ ] Add `positionEnd()` callback
- [ ] Add `accountSummary()` callback
- [ ] Add `heartbeat()` method
- [ ] Update `start()` to call `reqPositions()`
- [ ] Update `execDetails()` to call position tracking

E. Fix Config Inconsistencies

- [ ] Add `data_dir` to config
- [ ] Make `DRY_RUN` and `dry_run` both work (alias)
- [ ] Verify all env vars loaded correctly

📝 STEP 2: Update Existing Files

Files to Modify:

1. `src/ib_quant_kit/ib_client.py`

```python
# Add these imports at top
from .orders.lifecycle import order_lifecycle, set_ib_client
from .safety.circuit_breaker import get_circuit_breaker
from .costs.commission import cost_tracker

# In __init__:
def __init__(self):
    # ... existing code ...
    set_ib_client(self)  # Make this instance globally available

# Add methods from artifacts:
# - position()
# - positionEnd()
# - accountSummary()
# - heartbeat()
# - cancelOrder()
```

2. `src/ib_quant_kit/pnl.py`

- [ ] Replace entire file with complete PnL tracking artifact

3. `src/ib_quant_kit/config.py`

- [ ] Add `data_dir` field
- [ ] Add both `DRY_RUN` and `dry_run` (for compatibility)

4. `main.py`

- [ ] Replace with `production_main.py` artifact OR
- [ ] Keep simple version but add circuit breaker checks

🧪 STEP 3: Testing Phase (Week 1-2)

Day 1: Unit Tests

- [ ] Test circuit breaker all limits:

- Daily loss limit
- Drawdown limit
- Position exposure limit
- Order rate limit
- Connection health check

☐ Test commission calculations:
- Stock commissions
- Regulatory fees
- Slippage estimates

☐ Test order lifecycle:
- Order registration
- Status updates
- Timeout cancellation
- Bulk cancellation

☐ Test position tracking:
- Fill updates
- Average price calculation
- IB position reconciliation

Day 2-3: Integration Tests

☐ Connect to IB Paper Trading
☐ Verify positions download on startup
☐ Submit test order in DRY_RUN mode
☐ Verify order appears in lifecycle manager
☐ Wait for timeout, verify auto-cancel
☐ Manual cancel test
☐ Kill switch test (create file, verify trading stops)

Day 4-7: Paper Trading with DRY_RUN=1

☐ Run continuously for 3+ days
☐ Monitor all logs
☐ Verify no errors
☐ Check that risk limits would trigger correctly
☐ Verify staleness guard works
☐ Test reconnection after manual disconnect

Week 2: Paper Trading with DRY_RUN=0

⚠️ **Still in IB Paper Account, but orders will actually submit**

- [ ] Set DRY_RUN=0
- [ ] Start with 1 share position sizes
- [ ] Run for full trading day
- [ ] Verify fills tracked correctly
- [ ] Check commission tracking accurate
- [ ] Verify PnL matches IB's reported PnL
- [ ] Test circuit breaker with small loss
- [ ] Manually trigger kill switch
- [ ] Review all logs end-of-day

📊 STEP 4: Paper Trading Validation (Month 1)

Daily Checklist

- [ ] Start of day: Check positions match IB
- [ ] Monitor circuit breaker status hourly
- [ ] Review cost tracking at noon
- [ ] Check for any timed-out orders
- [ ] End of day: Full reconciliation
- [ ] Record metrics in trading journal

Weekly Review

- [ ] Calculate Sharpe ratio
- [ ] Analyze commission/slippage impact
- [ ] Review any breached limits
- [ ] Check for code errors/warnings
- [ ] Update risk parameters if needed
- [ ] Compare vs backtest expectations

Metrics to Track

Metric	Target	Actual	Status
Win Rate	>50%	___	___
Avg Win	>$50	___	___
Avg Loss	<$30	___	___

Metric	Target	Actual	Status
Profit Factor	>1.5	___	___
Commission %	<15%	___	___
Slippage (bps)	<5	___	___
Max Drawdown	<5%	___	___

🎯 STEP 5: Pre-Live Checklist (Month 2)

Only proceed if paper trading successful for 30+ days

Risk Management

- [ ] Circuit breakers triggered correctly in tests
- [ ] Kill switch tested and works instantly
- [ ] Position limits enforced
- [ ] Order rate limits enforced
- [ ] Commission tracking accurate to within $1
- [ ] Slippage estimates reasonable

Code Quality

- [ ] No errors in 30 days of paper trading
- [ ] All edge cases handled
- [ ] Reconnection logic tested
- [ ] Position reconciliation verified
- [ ] Order state tracking accurate

Monitoring & Alerts

- [ ] Slack alerts working
- [ ] Email alerts working (optional)
- [ ] Can monitor remotely (phone/laptop)
- [ ] Dashboard accessible
- [ ] Logs properly formatted

Documentation

- [ ] Emergency procedures written
- [ ] Trading journal started
- [ ] Risk parameters documented

- [ ] Know how to manually intervene via TWS
- [ ] Contact info for IB support ready

Financial Preparation

- [ ] Understand your tax obligations
- [ ] Account funded appropriately
- [ ] Backup funds available
- [ ] Written risk limits (dollar amounts)
- [ ] Exit strategy defined

🚀 STEP 6: Go Live (Month 3+)

Week 1: Micro Stakes

```bash
# Update .env for live
IB_PORT=7496  # Live TWS port (not 7497 paper)
DRY_RUN=0
MAX_POSITION_PER_SYMBOL=1  # 1 share only!
MAX_ORDERS_PER_DAY_PER_SYMBOL=10
MAX_DAILY_LOSS=100.0  # $100 max loss
```

- [ ] **Triple-check** connected to LIVE account
- [ ] Start with ONLY 1 share per position
- [ ] Trade for 5 days minimum
- [ ] Verify everything works identically to paper
- [ ] Review every single fill

Week 2-4: Gradual Scale

Only if Week 1 profitable or breakeven

- [ ] Increase to 5-10 shares per position
- [ ] Raise daily loss limit to $250
- [ ] Continue monitoring closely
- [ ] Still check every fill
- [ ] Daily journal entries

Month 2-3: Prove Consistency

- [ ] 30+ trades minimum
- [ ] Positive net PnL after costs
- [ ] Sharpe ratio >0.5
- [ ] No circuit breaker violations
- [ ] No manual interventions needed

Month 4+: Careful Scaling

Only if profitable for 3 months straight

- [ ] Scale position sizes by 50% each month
- [ ] Update risk limits proportionally
- [ ] Continue daily monitoring
- [ ] Monthly performance review
- [ ] Annual audit and tax prep

⚠️ RED FLAGS - STOP IMMEDIATELY IF:

1. **Net PnL negative for 5 consecutive days**
2. **Circuit breaker triggers more than once per week**
3. **Commission ratio exceeds 25% of gross profit**
4. **Any unexplained fills or positions**
5. **Code errors/crashes during trading hours**
6. **Slippage consistently >10 bps**
7. **Position drift detected (actual != expected)**
8. **Connection issues more than once per week**
9. **Can't explain any trade in your journal**
10. **Feeling stressed or emotional about trading**

When in doubt: FLATTEN AND STOP

📱 Emergency Contacts

Interactive Brokers Support: 1-877-442-2757

Your Broker Rep: _____

Backup Person (if you're unavailable): _____

Tax Accountant: _____

💾 Backup & Recovery

Daily Backups

- [ ] Export logs to external storage
- [ ] Backup .env file (encrypted)
- [ ] Save position snapshots
- [ ] Export PnL data

Disaster Recovery Plan

```bash
# If system crashes:
1. Access TWS directly via web/app
2. Check actual positions in TWS
3. Cancel all orders manually if needed
4. Close positions manually if needed
5. Don't restart algo until root cause found
```

📈 Success Criteria

After 6 months, you should have:

- [ ] Positive net PnL (after all costs)
- [ ] Sharpe ratio >1.0
- [ ] Max drawdown <10%
- [ ] 100+ trades executed
- [ ] Commission ratio <20%
- [ ] Zero unplanned interventions
- [ ] Complete trading journal
- [ ] Proven ability to scale

Only then consider increasing size significantly

🎓 Final Wisdom

Remember the hierarchy of goals:

1. **Don't lose money** (most important)
2. Don't lose a lot of money
3. Make a little money consistently
4. Scale profitable strategies carefully
5. Compound returns over time

The first year is about proving you can survive, not about getting rich.

Good luck and trade carefully! 🍀